

Same Trashbot, Different Streets: Local Legibility in Cross-Community Public Robot Deployment

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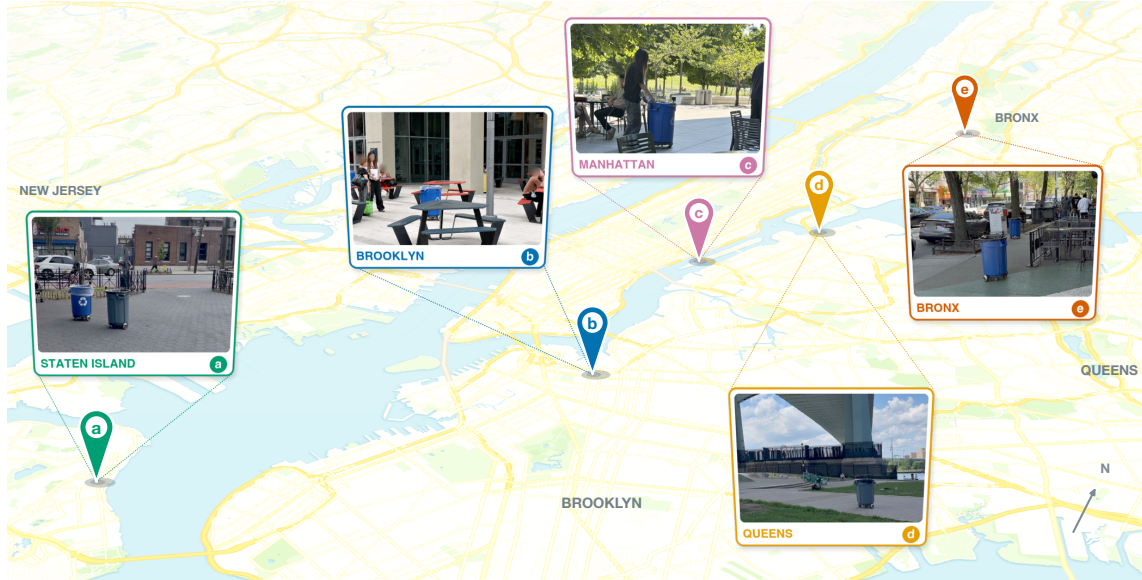


Fig. 1. Our study of how people perceived and interacted with a trashbarrel robot encompassed each of New York City’s five boroughs: (a) Staten Island, (b) Brooklyn, (c) Manhattan, (d) Queens, (e) the Bronx.

Public service robots are entering shared public space: they arrive unannounced and are rarely designed for the social life of the places they enter. To understand how people make sense of such robots, and how that differs by setting, we used a Wizard-of-Oz protocol to deploy a mobile trash barrel robot as a design probe across the five boroughs of New York City. We elicited behavioral interaction and performed post-interaction interviews to understand how people respond and make sense of these public robots. We find that people resolve the robot’s ambiguity by interpreting the robot’s intent through its functionality, and associating that functionality with the social and material order of the local site. What differs across sites is the local resources people draw on where they encountered the robot: firsthand knowledge of the neighborhood’s history and reputation, of how its space is used, along with cultural schemas of robots and surveillance. The apparent differences between neighborhoods are thus the output of one situated practice running on different local inputs. We argue this reframes public-robot sensemaking from an individual, concurrent cognitive act to a place-situated practice, where it points toward a solution in which robot that can answer the inquiry pattern in terms of each place people recognize. The goal is a middle path between deploying one universal robot everywhere and building a dedicated robot for each community.

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CCS Concepts: • **Human-centered computing** → **Participatory design**.

Additional Key Words and Phrases: human-robot interaction, public space, service robots, sensemaking, situated action, field deployment, local legibility

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1 INTRODUCTION

Autonomous robotic systems are moving out of the laboratory and into shared public space [24, 40]. Delivery robots now roam city sidewalks [45, 52], delivery drones ferry packages over residential neighborhoods [15], and driverless taxis carry passengers through downtown streets in cities around the world [8]. As these systems scale from isolated pilots toward routine presence, a tension has become increasingly visible: in February 2024, a crowd vandalized and set fire to a Waymo robotaxi in San Francisco’s Chinatown [48]. Public frustration has surfaced repeatedly, as residents complaining that delivery robots clutter street corners or block crosswalks [20, 52], and sometimes escalating into hostility and outright vandalism, with people kicking, tipping, or destroying robots they encounter on the street [3, 48]. These encounters raise questions about how robots enter the social life of a place: how do people understand what a robot is doing there, who is responsible for it, and whether its presence is appropriate?

HRI research has begun to document the work through which people accommodate and make sense of robots in public settings. Studies of delivery robots show how their movement depends on people adjusting their own trajectories and activities to grant them passage [14, 40]. In their study of trash barrel robots, Bu et al. [5] show that people interpret the broader circumstances of deployment, drawing on cultural knowledge, communal common ground, and personal experience to understand the robots and those behind them. Pelikan et al. [39] further identify local identity, physical environments, everyday activities, and sociability as central considerations for public robot design. This work establishes the importance of attending to the places in which encounters occur. What remains less clear is how the same robot comes to be understood differently across sites, and what those differences reveal about the interpretive work people perform when they encounter robots in their neighborhoods. Comparing interactions and perceptions across settings can help us understand what aspects of public robots are generalizable, and which vary by site; this in turn can help robot designers to better adapt robots for each deployment.

We pursue this comparison through a qualitative field study of mobile trashbarrel robots across six sites spanning all five boroughs of New York City. The sites differed in their everyday uses, sanitation arrangements, institutional affiliations, and degrees of public access. We deployed the robot in a riverside leisure park and a commercial corridor in the Bronx, two neighborhood parks in Queens and Staten Island, a university campus café in Manhattan, and an access-controlled industrial and commercial campus in Brooklyn (Figure 1). The study occurred across 24 field days, where the robot encountered approximately 491 people; we collected 89 post-encounter interviews (Table 1). The robot served as a common design *probe* across these settings: its familiar trash-can form offered a recognizable function, while its movement and onboard camera invited questions about its purpose, operation, and presence.

Our study addresses two research questions:

- **RQ1:** What do people seek to understand about a service robot when they encounter it in a public space?
- **RQ2:** how does this sense-making process differ from sites and why?

Our qualitative analysis examined participants' accounts within and across sites. Where interviews could be reliably linked to recorded encounters, we also examined how participants' subsequent explanations related to their observable conduct.

Our findings point to the importance of *local legibility* as a design objective: Local legibility supports the ability for citizens to assess a public robot in relation to the activities, responsibilities, and needs of the community. Framing the robot's role in the community helps people make sense of the robot's purpose, who is responsible for it, and its operational practices. Recurring concerns provide a basis for shared design principles to make robots more legible, but require local knowledge that should be enacted through institutional attribution, communication, behavior, and siting.

This work makes three contributions:

- We characterize how people make sense of public robots, identifying shared questions across deployment settings and the interpretive practices used to address them.
- A We show how people draw on local knowledge to address these shared questions, producing different judgments about the robots across deployment settings.
- We derive design implications for local legibility: making robots' purpose, institutional responsibility, and recording practices understandable within their settings, and adapting behavior and siting to local activities and needs.

2 RELATED WORK

2.1 Robots in Public and Urban Space

Robots have left the laboratory, and the field has followed them out. Jung and Hinds' call for in-the-wild work capable of supporting robust theory marks the turn [24], and the intervening years have supplied the deployments: sidewalk delivery robots operating at commercial scale [45], robotaxis carrying a quarter of a million paid rides a week [8], and drone delivery moving from waiver to routine certification [15].

Studies of these systems have documented how pedestrians negotiate passage with machines that do not negotiate back [40, 52], how robots and other sidewalk users compete for constrained space [19, 50], how passers-by are drawn into repairing a robot that has stalled [12], and how that competition falls unevenly on people with mobility disabilities [20]. A parallel line of field work has followed robots that stay in a setting long enough to become part of it, and has shown that what a robot means is shaped by the workflow, the organization and the people around it rather than by the machine alone [27, 28, 30]. Ours is the opposite arrangement: a robot that stays only hours, in six settings, so that what varies is the place rather than the length of the acquaintance.

A second strand documents what happens when the negotiation fails. Children obstruct and abuse social robots in malls, and asked why, give reasons that are about the robot's perceived agency and about the situation they are standing in [3, 36]; a robot left to move through an Italian city square is pushed, blocked and struck by passers-by [42]; a Waymo vehicle is set alight by a crowd in San Francisco [48]. Such incidents are typically narrated either as isolated pathology or as evidence of diffuse public hostility toward automation. Neither reading asks what the robot was taken to be at the moment it was attacked, which is the question we think does the explanatory work.

A third strand concerns the people a public robot records without recruiting. Scholtz's early taxonomy of HRI roles already separates the bystander—present in the robot's space, with no task relationship to it and no way to opt out—from the operator and the supervisor [43], and the privacy literature has since made that position its central problem: what a robot sees, who receives it, and on what terms it may be retained [11, 25, 29, 41]. Empirical work here tends to

find that concern tracks the framing and the setting rather than the sensor: people asked about a teleoperated home robot’s camera respond to how the capability is described [7], and bystanders of wearable cameras in public reason about the norms of the specific place they are standing in rather than about recording as such [10]. Nissenbaum’s contextual integrity gives that observation its general form—a flow of information is appropriate or not relative to the norms of the context it occurs in, not in the abstract [35].

Closest to our own framing is work that treats the city itself as the object of analysis. While et al. compare robotic experimentation across San Francisco, Tokyo and Dubai, showing that what a robot means—and what it is permitted to do—is constituted by local regulatory ambition, civic imaginary and governance arrangement [51]. We share the premise that place is constitutive rather than incidental. We differ in the level at which the comparison is drawn and in what the comparison is taken to show. Cross-city studies necessarily vary the robot, the operator, the regulatory regime and the population together, which makes place an aggregate independent variable and invites the conclusion that different publics receive robots differently. ~~Our design holds the machine, the operator and the protocol fixed and moves it across five settings inside a single municipal jurisdiction. What that isolates is not a smaller version of the city effect but a different object: not that publics differ, but that a single interpretive practice is running everywhere and drawing on whatever the immediate setting supplies.~~

2.2 Situated Action and Local Context

Pelikan’s argument for localism in robot design is the closest statement of the underlying commitment: public space is not a uniform substrate into which robots are inserted, and designing for it requires attending to the specific social organization of specific places [39]. We extend that commitment in two ways. First, we give it an empirical demonstration across six settings with the machine held constant, so that whatever varies cannot be attributed to the robot. Second, we draw a design consequence that localism alone does not entail. If the *questions* people ask are shared across places—who runs this, is it watching me, does it belong here—then attending to local specificity does not require a bespoke robot per community. It requires that the robot be able to answer a small, stable set of questions in terms the particular place makes sense of. That is the target we name local legibility (§6).

Legibility already has a technical meaning in HRI, and our use of the word is deliberately adjacent rather than identical. Dragan and colleagues define a legible motion as one from which an observer can rapidly infer the robot’s goal, and distinguish it from predictable motion, which matches what an observer expects once the goal is known [13]; the notion has since been carried through work on human-aware and socially aware navigation [26, 44]. That literature locates legibility in the robot’s trajectory and treats the inference problem as goal-directed: what is this machine about to do?

2.3 Sensemaking in and beyond HRI

We use *sensemaking* to describe the interpretive work through which people make an unfamiliar situation intelligible and decide how to respond. In HRI, this lens has largely focused on how individuals infer what a robot is, what it is doing, and how they should act from cues such as its appearance, motion, and behavior. This concern overlaps with work on robot legibility and predictability, which examines how people infer a robot’s goals or future actions from its movement [], and with research on mental models and expectation formation around autonomous systems []. Studies of bystanders and public-facing robots further show that interpretation occurs even outside explicit interaction, as people observe, accommodate, test, or otherwise respond to robots encountered in shared space []. Recent work has broadened this view: Bu et al. show that people draw on cultural knowledge, communal common ground, and

personal experience when interpreting public trash robots [5], while Pelikan et al. highlight local identity, physical environments, everyday activities, and sociability as consequential to public robot encounters [40]. Yet much of this work still centers the encounter itself, leaving less clear how knowledge of the surrounding place shapes interpretations of the same robot across different contexts. Beyond HRI, Weick’s formulation treats sensemaking as retrospective, social, and ecological: people interpret situations in relation to prior action, construct plausible accounts through socially available understandings, and draw on cues furnished by their environment [49]. Empirical HCI research has similarly examined how people construct and revise explanations when they encounter ambiguous technological behavior []. Such work shows that people do not interpret technologies from system behavior alone, but draw on prior experience, social knowledge, and contextual cues to develop plausible accounts of what is happening []. This broader view shifts attention from decoding the robot itself to understanding how interpretations emerge from the community in which it is encountered. We bring this perspective to public robots through a comparative study of the same robot across multiple settings, asking which interpretive concerns recur and how the local knowledge used to address them varies by place.

3 METHOD

3.1 The TrashBot as a Probe

A trash can is among the most ignorable objects in public space. People walk past it, use it without thought, and rarely treat it as something that requires interpretation. Yet when that same trash can is given wheels and made to move toward people on its own, it stops being background infrastructure and becomes a question. Passersby slow down, narrate it to companions, wonder who is behind it, and ask what it is doing here. The object is familiar; its behavior is strange. This tension—a recognizable everyday thing made unfamiliar through robotic movement—is what makes a mobile trash robot a productive probe for studying how people make sense of autonomous systems when those systems appear, unannounced, in the spaces of everyday life. We use *probe* in a specific sense. Cultural and technology probes are deployed to provoke responses that a direct question would not surface [18, 22]; breaching experiments make a taken-for-granted order visible by disturbing it, and are productive in design research for exactly that reason [9, 17]. A trash barrel that moves is closer to the second than the first. It does not ask people anything. It breaches the ordinary status of a fixed receptacle and thereby makes visible the work people were already doing to read the objects around them—work that is invisible while the barrel stays put. The practical demands of running such a probe repeatedly in public space are their own matter [4, 38], and we return to them in §7.

Prior work on trash barrel robots has shown that people readily engage with such systems and has examined how individuals interpret and respond to them in a single setting [2, 6]. Our TrashBot (Figure 2) follows the same trash-barrel form factor. This work has largely framed sensemaking as an individual, in-the-moment cognitive process: how a given person, encountering the robot, arrives at an understanding of what it is and what it wants [5]. We extend this line of work in two ways. First, we move from a single deployment site to six structurally distinct settings spanning all five boroughs of New York City. Second, and more consequentially, we reframe sensemaking itself—from a private cognitive act to a situated practice, one in which people interpret the robot using the social and material resources that the surrounding place makes available to them.

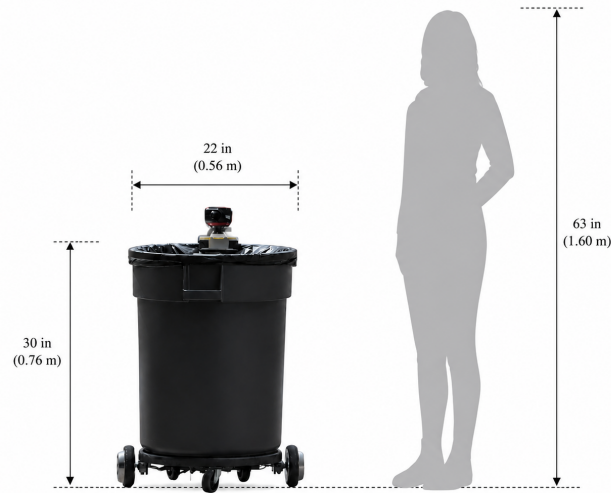


Fig. 2. TrashBot at scale

3.2 Deployment and Data Collection

New York City is a compact setting in which one robot can be exposed to many local orders. Its five boroughs share a single municipal government and a single citywide sanitation agency [33], yet differ significantly in population, culture, and income level: borough populations run from about half a million in Staten Island to 2.7 million in Brooklyn [31]; median household income from under \$50,000 in the Bronx to over \$100,000 in Manhattan [37]; and the foreign-born share of residents from a quarter in Staten Island to nearly half in Queens [16, 32]. Moving the same robot between boroughs therefore varies the place it appears in while holding the rules it operates under constant.

We conducted a multi-site field study with the same TrashBot across public and quasi-public settings in New York City's five boroughs. The study used a Wizard-of-Oz setup throughout [2, 5, 6]. To people encountering the robot, TrashBot appeared to move autonomously; in practice, a researcher controlled its movement remotely from outside participants' view. Keeping the robot and operating arrangement consistent allowed us to examine how encounters with the same system were understood in different local settings.

TrashBot consisted of a trash receptacle mounted on a hoverboard base, allowing it to move and rotate in place. An onboard camera recorded the interaction space from the robot's point of view. During deployment, the operator controlled TrashBot through the site: it approached people, stopped near them for possible interaction, and patrolled the surrounding testing area. People could dispose of waste, approach or avoid the robot, speak or gesture toward it, or simply observe it. We considered an encounter to have occurred when a person visibly oriented toward TrashBot, so direct disposal was one form of encounter rather than a requirement for inclusion.

Recruitment took place after the encounter had run its course. Once a person's interaction with TrashBot ended, a researcher approached them and invited them to take part in a short post-encounter interview. The approved protocol used this ordering because requesting consent before the interaction would reveal the research context and could prime the participant's response to the robot. The interviews were semi-structured around participants' interpretations of TrashBot. Researchers asked about their immediate impressions of the robot, how they understood the idea

Table 1. TrashBot deployment sites and interview corpus.

Borough	Site	Setting	Dates	Field days	Interview days	Interviews
Staten Island	Tappen Park	Neighborhood park	Jul 2025	4	4	10
Queens	Astoria Park	Riverside leisure park	Jul 2025	4	4	13
Bronx	Little Italy	Commercial corridor	Apr–Oct 2025	7	7	33
Bronx	Starlight Park	Neighborhood park	Sep 2025	2	1	1
Manhattan	Roosevelt Island café	Campus café	Jul 2026	3	3	17
Brooklyn	Brooklyn Navy Yard	Industrial / commercial campus	Jul–Aug 2026	4	2	15
Total				24	21	89

Deployment days count all field days at a site, including days on which no interviews were conducted. Interview days count field days that produced at least one interview record. Interviews count interview records rather than unique individuals; several records correspond to pairs or small groups interviewed together.

of an autonomous trash robot, and what they thought about deploying a system like TrashBot in their community or surrounding setting. Follow-up questions probed what participants found useful, concerning, confusing, or inappropriate and where they thought such a system would or would not fit. Most recorded interviews lasted approximately three to five minutes, although the approved protocol allowed for a brief interview of up to approximately ten minutes. Interviews were video-recorded when participants permitted video recording. When a participant declined video but permitted audio recording, the interview could be recorded as audio only.

Question wording and follow-up probes varied across researchers and sessions. The July 23 Roosevelt Island interviews included two direct questions that were not part of the 2025 interviews: how participants felt about the robot specifically in that campus setting and how they felt about autonomous robots more generally. Place-directed discussion from that session was therefore partly elicited rather than arising only from participants' spontaneous accounts.

Table 1 and Figure 3 summarize the deployment sites and interview corpus. Across the six sites, TrashBot was deployed on 24 field days, 21 of which produced interviews. The current corpus contains 89 interview records. Because several interviews involved pairs or small groups, these records do not correspond to 89 unique individuals. We therefore report interview records rather than reconstructing a person-level sample size from the recordings. Interviewees included people using the sites for leisure, work, or everyday activities, as well as people involved in local sanitation or maintenance.

Field days and interview days do not always coincide because not every deployment day was used for interview recruitment. Some field days were used for community engagement rather than collecting interviews, and these days remain part of the deployment record even when they produced no interview records.

The study was approved by our institution's Institutional Review Board for Human Participants (protocol number omitted for anonymous review). The protocol uses an alteration of informed consent because obtaining consent before the encounter would disclose the study and could shape participants' initial responses. Participants were therefore approached for consent after their interaction with TrashBot. The researcher then introduced the study, provided the approved alteration-of-consent form, and requested permission for the interview and recording.

The approved procedure waived written documentation of consent rather than documentation of consent itself. Participants were to receive a copy of the consent form, review it, and keep it for their records; no signature was

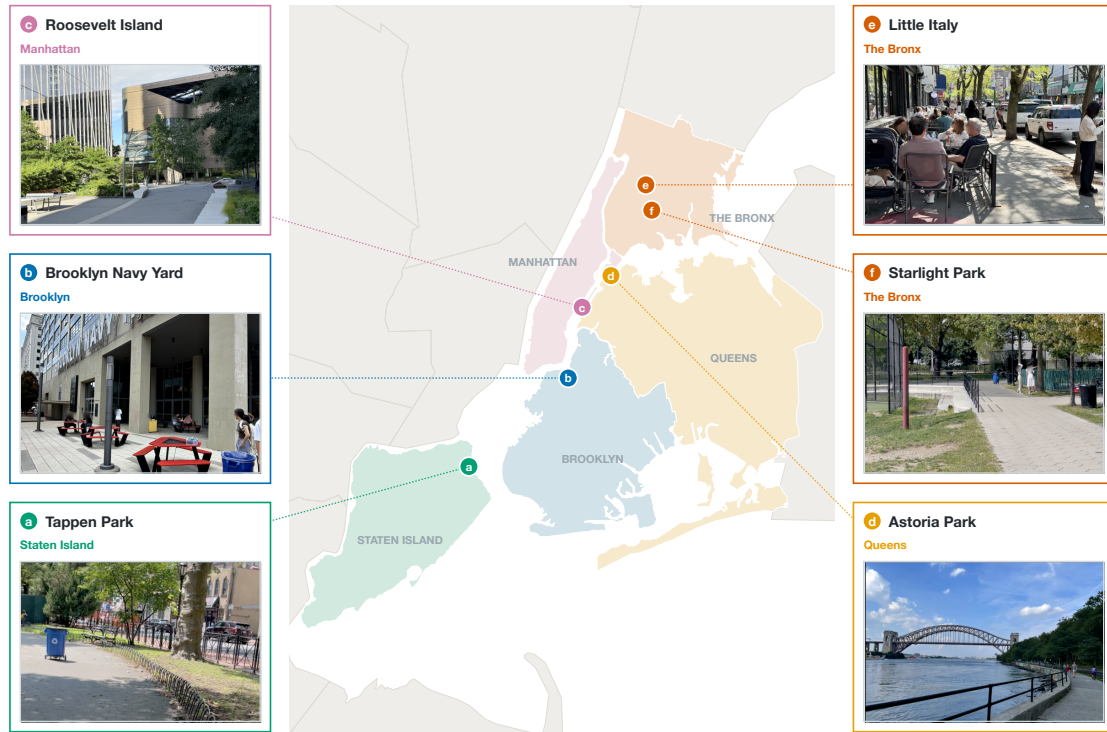


Fig. 3. The six TrashBot deployment sites across New York City. The callouts show Tappen Park in Staten Island, Brooklyn Navy Yard in Brooklyn, the café on Roosevelt Island in Manhattan, Astoria Park in Queens, Little Italy and Starlight Park in the Bronx as encountered during the field deployments.

required. Consent was documented through the recorded interaction with the researcher. This record was particularly relevant because identifiable video could be retained for analysis and publication. The protocol also specified that no signatures would be collected and that participants' permission would instead be recorded in audio and/or video.

The 2025 fieldwork included documented departures from this approved consent procedure. Across 48 interviews collected during the Staten Island, Queens, and Bronx deployments, 31 participants were not given the alteration-of-consent form, and 17 reviewed it but did not keep a copy. Verbal permission to conduct and record the interview was requested and recorded in every case. These departures were reported to the IRB, which determined them to be non-serious, non-continuing non-compliance presenting no unanticipated risk to participants, and approved retention and use of the data.

3.3 Sites

Tappen Park (Staten Island) is a neighborhood park in Stapleton, near a commercial district, a few blocks away from the waterfront of Upper New York Bay. It functions as a gathering place for the surrounding community, where most visitors live nearby, many are acquainted with one another. The park is used more for sitting, talking, and passing time than for organized recreation. It is a medium size park, and fixed trash barrels are sparsely located around corners and

the entrance of the park. The surrounding area is historically under-invested, and its reputation is documented rather than merely local: the Richmond County District Attorney’s fentanyl task force identifies Tappen Park and Stapleton as one of only two open-air drug markets on Staten Island [46], and the police precinct covering Stapleton and the rest of the North Shore accounts for the large majority of the borough’s gun violence—21 of 27 borough-wide shooting incidents in 2023 and 9 of 11 in 2025 [34]. Gun violence in the precinct fell over that period, though the counts are small and the precinct covers the whole North Shore rather than Stapleton alone. We deployed the robots around the open lawn, with the wizard seated on a bench nearby.

Astoria Park (Queens) is a large riverside leisure park on the East River, used for running, cycling, family outing, sunbathing, and extended outdoor recreation. We deployed the robot around its recreational leisure area, including the public swimming pool entrance, along the riverside walk, and beside the skateboard park, which placed them among parents bringing their children to the pool, cyclists pausing to rest, and people relaxing on the lawns and benches. As at Tappen Park, fixed trash cans are sparsely distributed, with considerable walking distance between them. This site was covered by the same Parks Department permit. Wizards were seated on a bench at the edge of the deployment area.

Most Bronx interviews were collected in **Little Italy**, a dense commercial corridor centered on Arthur Avenue in Belmont and historically identified with the borough’s Italian immigrant community [1, 21]. The corridor concentrates restaurants, shops, market activity, and heavy pedestrian traffic. Sanitation was already visibly organized through local cleaning crews and porters, and fixed trash barrels were set across the street. Deployments were conducted in the trapezoidal plaza outside a pizzeria on the corridor. Access was arranged through the executive director of the neighborhood’s business association, who introduced us to the surrounding business owners. A smaller Bronx deployment took place at **Starlight Park**, a urban park located along the Bronx river. We reached out to a park administrator

The Manhattan deployment took place at a café beside a research institute on **Roosevelt Island**, a residential island in the borough of Manhattan with roughly 20,000 residents. We deployed around the café and its open seating area, where tourists, people affiliated with the institute, and local residents gather, and recruited from among the people sitting there. Access was granted by the institute’s campus safety department.

The Brooklyn deployment took place at the **Brooklyn Navy Yard**, an access-controlled industrial and commercial campus used largely by people working at organizations located there. Access came through the chair of the campus’s community-engagement group, who invited us to bring the robot to a community event at which residents and workers were cleaning tree pits. The Brooklyn deployment therefore differs from the others in one respect worth stating: the robot arrived as an invited participant in a neighborhood activity rather than as an unexplained object, and we read the accounts collected there with that framing in view.

These sites are not matched conditions. Borough, setting type, local population, degree of public access, existing sanitation infrastructure, and institutional context covary across locations, so the study cannot separate the effect of one factor from another. We therefore do not interpret site as an independent variable or use the data to estimate neighborhood-level differences in acceptance or behavior. Site instead provides context for interpreting what participants believed the robot was, who they thought might be responsible for it, what risks they associated with it, and whether it fit the surrounding environment.

4 ANALYSIS

Our analysis draws on two records of the field encounters: what participants said in post-encounter interviews and what TrashBot’s onboard camera recorded while interactions were unfolding. We first worked with the interview and video materials separately because they capture different moments and support different kinds of claims. We then linked the two when the same participant could be identified reliably across both records. This organization allows us to examine stated interpretations, situated conduct, and the relationship between them without treating the two sources as interchangeable.

Researchers conducted the substantive analysis of both sources. Automatic speech recognition was used to produce working interview transcripts, but researchers reviewed the interviews, organized the thematic material, reviewed the onboard video, and established participant-level links between interview and video records. These analytic judgments were made from the source recordings and accompanying field records rather than produced through automated coding or identification.

4.1 Interview Analysis

Researchers manually reviewed the interview recordings and working transcripts. The corpus was divided across several researchers, who took primary responsibility for material from different deployment sites. For each interview, the researcher prepared or reviewed a transcript, summarized the participant’s account, extracted passages relevant to the research questions, and recorded concise observations and thematic keywords in a site-level analysis deck. The current decks organize material at the participant level and retain the interview account together with source-file information, selected transcript passages, analytic notes, thematic labels, and corresponding onboard-camera evidence when a reliable match is available. The keywords functioned as working summaries for organizing the material rather than as a predefined coding scheme, and participants were not assigned to mutually exclusive categories.

Working transcripts were generated with the Whisperdistil-large-v3 automatic speech recognition model with word-level timestamps, which processed locally. We used these transcripts as an index into the recordings rather than as an authoritative verbatim record. The analysis decks distinguish verbatim transcript passages from researchers’ reported speech and interpretation. Whenever the paper reports a participant’s words as a direct quotation, a researcher checks the passage against the original audio or video. The broader working corpus may therefore contain uncorrected machine transcription, while quotations used as evidence in the manuscript are source-verified.

Some records required additional caution. Automatic transcription was weak in portions of several interviews, some interviews involved translation, and a small number of recordings were incomplete or truncated. These records remain documented in the analysis materials, but uncertain wording is not treated as a verified quotation. Where a translated statement rather than the participant’s original-language speech appears in the working record, we do not present that translation as a verbatim quotation without checking the underlying recording.

Analysis proceeded from participant-level summaries to comparison within and across sites. The first pass stayed close to each interview, recording the concerns, comparisons, assumptions, and proposals that participants themselves raised. Researchers noted how participants understood the robot’s function, what they noticed about its movement or camera, what risks or uses they associated with it, and what existing technologies, institutions, or local conditions they used to explain their judgments. Cross-case comparison then identified recurring ways of accounting for the robot. The analytic groupings used in the Findings developed through this comparison rather than being assigned to the interviews in advance.

Four forms of reasoning recur in the present synthesis. *Institutional* accounts concern who operates, owns, authorizes, or might receive data from the robot. *Threat* accounts concern what people might do to the robot or what risks might follow from its presence, including theft, vandalism, physical safety, and possible effects of automation. *Body and privacy* accounts concern the camera, recording, data access, consent, physical proximity, and the bodies or activities the robot can see. *Infrastructural-fit* accounts compare TrashBot with existing receptacles, sanitation labor, pedestrian conditions, maintenance practices, and everyday routines. These categories organize recurring resources in participants' accounts. A single interview may draw on several of them, and the categories do not define participant types.

We also retained comparisons that participants introduced themselves when making sense of the robot. Participants drew on other robots, consumer technologies, sanitation systems, institutional actors, media narratives, and prior experiences with cameras or data collection. These comparisons were analyzed within the accounts in which they appeared rather than converted into a separate coding scheme. They are especially useful when similar features of TrashBot receive different interpretations across sites because participants draw on different prior experiences or local reference points.

4.2 Video Analysis

Researchers manually reviewed each available onboard-camera recording in full rather than sampling only clips linked to interviews. The review served two purposes. It allowed us to locate the encounter associated with an interviewed participant when a reliable correspondence could be established, and it preserved interactions that were analytically relevant even when the person was not later interviewed. The latter material provides a broader record of how TrashBot was noticed, approached, used, ignored, tested, or discussed in the field.

We treated the footage as interactional material rather than as a corpus of behavioral variables to be exhaustively coded and counted [23]. During review, researchers attended to observable conduct such as distance from the robot, approach or avoidance, hesitation or testing, disposal behavior, orientation to companions, and speech or gesture directed toward the robot. Reviewers recorded these observations at the level of specific episodes and retained source files and timestamps where available. These observations are used to describe and interpret encounters rather than to support statistical comparisons between sites.

Onboard-camera coverage was not uniform across deployments. Some sessions have footage that can be searched for interview encounters, while others have incomplete, unavailable, or not yet processed robot-POV material. One of the three Roosevelt Island sessions has no onboard recording, and Brooklyn Navy Yard footage still requires processing before participant-level matching can be completed. Interviews from sessions without usable onboard footage remain part of the interview corpus but cannot support participant-level claims about conduct during the preceding encounter.

For participants who were interviewed, we searched the onboard recording from the corresponding session for the interaction that preceded recruitment. Matching relied on temporal correspondence and the sequence of events within the session together with visual and situational information available to the research team, including appearance, clothing, posture, companions, and position in the scene. Reviewers also considered the transition from the encounter to recruitment when it was visible in the records. The match was therefore an analyst judgment based on converging contextual cues rather than an automated identification procedure. We included a case in the paired analysis only when the correspondence could be established with sufficient confidence.

This linkage remained internal to the research process. Participants were represented in the analytic materials and in the paper through study IDs rather than names. The purpose of matching was to connect two records of the same

encounter, not to establish or report a participant’s external identity. Interviews that cannot be linked reliably to onboard footage remain part of the interview analysis, and interactions visible in onboard footage can contribute to the video analysis without supporting claims about what that specific person later thought or said.

4.3 Bridging Conduct and Account

Interview and onboard-video records describe different parts of the same encounter. The video records conduct as the interaction unfolds, before the participant has been asked to explain what happened. The interview is produced afterward and in response to a researcher’s questions. We therefore do not treat the two records as parallel measures of a single attitude or use one to validate the other. Instead, the paired analysis examines how a participant’s later account relates to conduct recorded during the encounter [47].

For cases with a reliable linkage, the unit of analysis is the *encounter-plus-account*: a participant’s recorded interaction with TrashBot considered together with that participant’s post-encounter interview. The matching procedure described in §4.2 establishes which records can enter this analysis. Interviews without a reliable onboard match remain part of the interview analysis, and onboard encounters without a corresponding interview remain part of the video analysis. We make participant-level claims across the two sources only when the linkage is sufficiently confident and both records are permitted for use under the final consent determination.

For each paired case, we first considered the two records separately. The conduct record focused on what was observable during the encounter, including approach or avoidance, distance from the robot, hesitation or testing, disposal behavior, orientation to companions, and speech or gesture directed toward TrashBot. The interview record focused on how the participant subsequently explained the robot, including the concerns, assumptions, comparisons, and local references used to account for the encounter. Keeping the records separate at this stage helped avoid reading the later explanation back into an action whose meaning was not evident from the video alone.

We then examined how the two records related. In some cases, the interview provides an explanation that is consistent with conduct visible in the footage. In others, it introduces considerations that could not be observed directly, such as assumptions about who operates the robot, concerns about recording or data use, or judgments about whether the robot fits the surrounding setting. The interview can also clarify conduct that is ambiguous on video or show that the participant’s interpretation changed during the encounter or after receiving additional information. Differences between conduct and account are therefore treated as part of the case rather than as evidence that one source is inaccurate.

Paired cases were analyzed within the site-level workflow, with interpretations checked against the source recordings and reviewed within the research team before use in the manuscript. The comparison remained qualitative and case based. We did not assign participants to mutually exclusive behavioral types or use relationships between conduct and account to estimate differences between sites. Paired cases are used in the Findings when the two records together provide information that either source alone cannot support.

This structure also sets a boundary on what can be inferred from unmatched material. An interview can support claims about what a participant said or how they later interpreted TrashBot without supporting a claim about their earlier conduct. An onboard encounter can support a description of observable behavior without supporting a claim about what that person believed. The paired analysis is reserved for cases in which both records can be connected to the same encounter.

5 FINDINGS

We report the findings in three parts. Section 5.1 gives an empirical account of the encounters at each site, including the cases in which an interview could be linked to the participant’s recorded conduct. Section 5.2 describes what people considered when they interpreted the robot, which recurred across sites (RQ1). Section 5.3 compares how those interpretations differed from site to site and what local knowledge the differences drew on (RQ2).

5.1 Encounters at the Six Sites

At every site for which we have reviewed onboard footage (Staten Island, Queens, and the Bronx), most people who noticed the robot treated it as a trash can: they walked up to it and put rubbish in, without instruction, prior familiarity with robots, or a shared language with the researchers. The accounts they gave afterwards differed by site. Institutional attribution also differed between the park and street sites and the two campus sites. In a park or on a commercial street the organization behind an unfamiliar robot was not apparent, whereas at the research-institute café and the Navy Yard the setting itself suggested who might be responsible. **TODO: Albert/Xiang: expand each site below into a full account (who approached, what people did on camera, what they said). Site order follows Table 1. Kinds, not counts; no comparison of rates across sites.**

Tappen Park (Staten Island). People on the benches used and played with the robot. A man held out his drink can to it as it passed (Figure 4a, 1 Jul). On 7 Jul another looked for something to throw away, found nothing, finished his drink, and put the bottle in (Figure 5c–d). He described the sequence himself, “*everything there is trash, and that’s my present, so ...I’ve looked around to see if there was trash there*”, and said he did it to try the robot: “*it’s like a new experience, like I really want to ...try with this*”. On the same afternoon the robot’s camera recorded the only refusal in the corpus (Figure effig:vignettese–f). A participant first joked with the robot, naming it R2D2, then objected to its camera, “*don’t video us ...you’re staring at me with a camera like this*”, knocked it over, kept objecting while the camera lay on the ground (“*take it somewhere else*”, “*I don’t like my picture*”), and then apologized to it: “*I’m sorry, buddy ...I didn’t mean to hurt you.*” In the interview he tied the objection to the place, “*everything is under the eye of the sky these days, and it takes away a sense of freedom*”, asked the robot to “*please note that in your memory*”, and summarized the episode as “*that’s why it turned from cute to offensive*”. Another participant was at first unsure whether he was being recorded, “*I was wondering if you guys were recording me*”, decided that the researchers were driving the robot, “*I realized that you guys were operating it ...so I wasn’t scared*”, and kept that reading when told it was autonomous: “*I thought you guys directed it to me*”. Some participants situated the robot against the park’s reputation, including concerns about drugs and public safety and how unfamiliar equipment would be treated there; one recommended operating in daylight only.

Astoria Park (Queens). People used the robot quickly and without comment. A passerby walked straight to the robot, dropped waste in, and left; a boy who noticed it looked around for something to throw away; two participants raised their phones and filmed it as it approached (Figure 4b, 17 Jul). On 15 Jul a man stopped in front of the barrel with a sealed bag in hand, asked the research team whether he was allowed to use it, and then disposed of the bag (Figure 5a–b). Through an interpreter he explained that the machine looked like a prototype that might not yet take rubbish. This is the only disposal the onboard camera records from an empty liner to the object at rest. On 16 Jul a participant screamed and moved away each time the robot approached, kept watching from a distance (Figure ??), and then gave a measured interview: people would steal it (“*I think people will just take them*”), a municipal version would read as surveillance



Fig. 4. Two ordinary responses, from the robot’s cameras (faces obscured). (a) A man on a bench holds out his drink can to the robot as it draws level with him, from the forward-facing camera used in the first session. (b) A participant raises his phone and films the robot as it approaches; the research team is on the bench behind him.

the way the police robot dogs did, a park suits it better than a street, and “*at least it’s kind of friendly, right? Doesn’t have any weapons*”. The lawns at Astoria Park are used for sunbathing, and the privacy concern there was about bodies: “*...people with bikini ...don’t want to be filmed*”. One participant asked that the camera be made clearly visible and that the robot keep about five feet away; another set six feet and grounded it in hygiene. Asked who was behind the robot, one participant assumed it was the city’s and was reassured, another distinguished a university experiment from what a municipal version would mean, and a third read it as robotic policing. Because fixed bins are sparse, participants judged the robot useful, especially on crowded weekends. In the permission case and in a second Astoria encounter, using and watching the robot did not require a shared language with the researchers; only the interview afterwards did.

Little Italy, Arthur Avenue (Bronx). The robot moved among the sidewalk tables outside a pizzeria and the street beside the municipal bins (Figure 6). On 29 Sep two people at a café table reached across its rim in turn as it stopped beside them (Figure 6a). The participant at that table then argued in the interview that the robot did not belong on the block: too much foot traffic, no good in the snow, better suited to an enclosed space. Cleaning was already visible on the block: porters, the landlords’ own arrangements, and a public-safety officer on sanitation duty who looked the robot over (Figure 6b). One participant explained that “*the community already have their porters that clean here every single day*” and redirected the robot to 125th Street, 149th, and Kingsbridge, streets he considered dirty; a merchant called it “*useless*” because landlords handle the bins building by building; students objected that “*there’s three trash cans on the street*”. Another participant asked where we would be permitted to operate and noted that street cleaning is already tied to the sanitation department and to the business associations. Theft came up repeatedly: “*they will take anything to make money to buy drugs*”; “*Here, no way. No way. You got to bring it inside with the remote control.*”; “*Anything is not tied down over here*” (Figure 6c); “*People who steal here is the ones who live here*”; “*Eventually it’ll work here once the crime rate goes away*”. On the camera, one participant distinguished the street from the home: “*But you ain’t in your house, you’re on the streets*”. Another came over alarmed at the camera, “*oh, what the hell?*”, and after learning that a person was driving the robot re-described it as a game, “*It’s like playing a PS spot.*” (The disclosure departed from our protocol; see Section 7.) A sanitation worker interviewed on her route moved from “*My only, like, question is, does that have an*



Fig. 5. The three encounters of Section 5.1, seen from the camera on the barrel’s rim (faces obscured; times are within the session recording). (a) The Astoria Park participant stops in front of the barrel with a sealed bag in hand; he has just asked the research team whether he is allowed to use it. (b) Three seconds later the bag goes in—the one disposal in the corpus that the onboard camera records from an empty liner to the object at rest. (c) At Tappen Park a man on a bench, a part-finished bottle of cola beside him, watches the barrel arrive. (d) Forty seconds later he reaches across the rim as the barrel turns toward him; the bottle, clear plastic against a black liner, is not resolvable in the frame. (e) The same afternoon, another participant stands over the barrel and addresses its camera directly (*“you’re staring at me with a camera like this”*). (f) Six seconds later, after he has knocked the barrel over: the camera now lies on the ground beside its own rim.



Fig. 6. One block, three resources: Arthur Avenue, 29 September, from the camera on the barrel’s rim (faces obscured; times within the session recording). (a) The robot stops at a pizzeria’s sidewalk table and the two people eating there reach across its rim in turn—the participant whose interview then explained that it did not belong on the block (§5.1). (b) Three minutes later it stands beside a municipal litter basket while a public-safety officer on sanitation duty looks it over: the labor the block’s residents measured it against. (c) The participant who told us that *“anything is not tied down over here”* inspects it after being called across the sidewalk.

effect over what I do?” to *“It looks like it makes people job, my job easier”*, and then to the worry that her employer would conclude that *“we won’t need that many workers”*.

Starlight Park (Bronx). The single interview collected there connected the robot to the surrounding Spanish-speaking, Latino, and Caribbean community, in terms of bilingual communication, culturally recognizable design, and how the camera would be read: *“in this neighborhood, I feel like a camera is going to make folks feel like, who’s watching? You know? Is this the police, you know? ...Or even, is this ICE?”*

Roosevelt Island café (Manhattan). This is a campus setting, and many users had an existing relationship with the institution responsible for the space. Some had seen TrashBot before or related it to other technical projects on campus, and read it as a student project. Participants reasoned about recording as a matter of procedure and consent, noting that a machine moving through a space where people have not consented to being filmed raises a question of permission, and that audio and video collection ought to be consented to. **TODO: Add two or three quotes from the Roosevelt Island transcripts (student project; consent). Onboard footage for this site has not been reviewed, so there are no linked cases yet.**

Brooklyn Navy Yard (Brooklyn). The robot arrived as an invited participant in a community event (cleaning tree pits), and the people we spoke to were there to work. Their accounts distinguished the robot's safety inside the controlled campus from anticipated risks on the streets beyond it, and none of them raised theft, stripping, vandalism, or tampering. **TODO: Reconcile the count: the earlier draft said eight participants; Table 1 lists 15 interview records.** On the camera, consent was the test, "*if it's like, you know, unconsentually, that's a bit, like, weird*", glossed as "*major privacy*". One participant saw the camera as part of what a robot is, "*the camera is because it's robot, right? It cannot be without camera, right?*", and another approved of the recording because it would show how much rubbish is actually thrown away (paraphrased; see the quote policy below). **TODO: Add one or two quotes on usefulness at the Navy Yard. Onboard footage not yet reviewed.**

Linked encounters: what people did and what they said. For seven encounters we hold both the robot's recording of what a participant did and that participant's later account (Table ??). In most of them the account matched the conduct or supplied the reasoning behind it, as in the Astoria permission case, the Tappen trial, and the Tappen refusal. In two cases the two diverged, in opposite directions: use followed by an objection to the deployment at the Arthur Avenue table, and avoidance followed by a considered appraisal at Astoria Park on 16 Jul. In three cases the reading changed during the encounter after the participant learned or decided who was behind the robot: a researcher disclosing that a person was driving it (Little Italy), the participant's own inference (Tappen Park), or a camera that kept looking while the question stayed open (the Tappen refusal). Using the robot therefore did not entail endorsing its presence, and the accounts drew on concerns, such as theft, surveillance, sidewalk width, and winter, that the recorded conduct never showed. Onboard footage has been reviewed for Staten Island, Queens, and the Bronx only; the rim of the barrel is in frame for part of each recording; and clear plastic leaves no trace against the black liner. We therefore report kinds of relationships, not rates, and make no comparison of their frequency between sites.

5.2 The Shared Interpretation Process

TODO: RQ1. Each paragraph below should grow into a short account with one or two quotes; the evidence is in Section 5.1. Do not reuse the same quote twice. People interpreted the robot's form and function first. They recognized it as a trash can and used it before settling any other question about it (Section 5.1). The question that mattered most was who was behind the robot and who was responsible for it. Candidate answers came from the setting: the city, a university or research project, the police or ICE, a business association, or a person with a controller. Once the question was answered, by a researcher's disclosure (Little Italy) or by the participant's own inference (Tappen Park), the participant stopped worrying about the robot. When it stayed open while the camera kept recording, the encounter ended in refusal (Tappen Park). Participants also asked whether the robot was recording them and what would happen to the footage. The question was raised during encounters, addressed to the camera, and again in interviews. What the camera meant depended on what the place is used for and on who was assumed to be watching (Section 5.3).

Participants most often evaluated whether the robot’s work was needed, judging whether the job was already being done by the sanitation arrangements they knew. Participants also asked whether the robot belonged in the place and what would happen to it there, applying the reputation of the place to the robot’s survival (theft, stripping) and to its fit (foot traffic, winter, better indoors). Interpretation continued after the encounter. Interviews supplied reasoning the footage could not show, and the account was produced under different demands from the conduct (Table ??).

5.3 How Interpretations Differed Across Sites

TODO: RQ2. Organize by the question, compare sites within each paragraph, and end each with what recurred and what varied. Site descriptions stay in Section 3.3; only the readings go here.

Who runs it. At Starlight Park the candidates were the police or ICE. On Arthur Avenue participants pointed to jurisdiction, the sanitation department, and the business associations. At Astoria Park one participant assumed the city and was reassured, another distinguished a university experiment from a municipal version, and a third read the robot as robotic policing. At Roosevelt Island it was a student project, and at the Navy Yard participants did not question who was responsible. The same question was answered from what people knew about who operates things where they live.

What the camera is doing. At Astoria Park the camera meant bodies on the lawn (“*people with bikini*”), requests for distance, and filming back. At Starlight Park it meant watching by the police or immigration enforcement. On Arthur Avenue the street was not the home. At Roosevelt Island and the Navy Yard the camera was a matter of procedure and consent (“*unconsentually ...weird*”), part of what a robot is, or a way to measure rubbish.

Whether the work is needed. On Arthur Avenue, porters, landlords, and three bins on the street made the robot redundant, and participants sent it to dirtier streets; the sanitation worker judged it against her own job. At Astoria Park, where bins are sparse, the robot was useful, especially on weekends. Participants judged the same robot differently because the sanitation infrastructure at each site differed.

What will happen to it. On Arthur Avenue the robot would be stripped for parts and should be brought inside, because “*Anything is not tied down over here*”. At Tappen Park the park’s drug history led to a recommendation of daylight only. At Astoria Park on 16 Jul, “*people will just take them*”. At the Navy Yard no participant raised the question; safety inside the campus was set against the streets beyond. The theft reading appeared where participants had a story about what happens to unattended property and did not appear where they had none.

What recurred and what varied. The questions recurred at every site. The knowledge used to answer them was local: the neighborhood’s history and reputation, how its space is used, its sanitation labor, and the institutions understood to operate there. **TODO: Two or three sentences that hand over to Section 6; no theory here.**

6 DISCUSSION

6.1 From Individual Cognition to Situated Practice

The reframing we have argued for is small to state and consequential to adopt. Public-robot sensemaking has largely been studied as something that happens inside a person during an encounter: cues go in, a mental model comes out, and what varies between people is what they brought with them. On our evidence the encounter is not the boundary of the process. The materials people reason with are lying around the site—who cleans this block, what this park is used

for, what happens to unattended property here—and the account is assembled from them at the moment of accounting, which is usually after the conduct it is supposed to explain. Three things follow for how the field works. First, *the unit of analysis should include the setting*, not as a moderating variable but as the supply of premises. A study that records only the person and the robot has thrown away the half of the system that determines the answer. Second, *disagreement between sites is not evidence that populations differ*. Our participants reasoned identically and concluded oppositely, and any design that reads the conclusions as attitudes will mislocate the cause—and, in practice, will licence claims about which communities are receptive to robots that our data do not support. Third, *what people say and what they do should be collected separately and compared, not averaged*. The gap between them is where the situated character of the process becomes visible; a method that treats the interview as a measurement of a disposition will record the gap as measurement error and discard the finding. The reframing also recovers something. Weick’s account of sensemaking was already social, retrospective and ecological (§2.2); HRI’s uptake narrowed it to individual cognition. We are not proposing a new construct so much as declining to keep the narrowed one.

6.2 Local Legibility

Legibility in HRI usually names a property of a robot: whether its intent can be read from its motion, whether its internal state is visible from outside. We mean something adjacent but not the same. A robot is *locally legible* when the people in a particular place can answer, from where they are standing, the questions that place leads them to ask about it. The distinction matters because the two can come apart completely. A machine can be perfectly legible in the HRI sense—smooth, predictable, its next move obvious—and remain unreadable in ours, because the question the street is asking is not *where is it going* but *whose is it*. Our participants were rarely confused about what the robot was doing. They were unsure who had put it there, and what the camera on it meant in this neighborhood, and whether the work it was doing was work that needed doing here. Those are the questions legibility has to satisfy, and none of them is answered by better motion planning. This gives a target that avoids two bad options. The first is uniform deployment: one robot, one label, one set of behaviors, everywhere. It fails not because places differ in taste but because a fixed answer addresses whichever question the designers imagined, and the question actually being asked varies. A prominent “CORNELL RESEARCH” label resolves a great deal on a university campus and very little on a block where the live question is whether the thing is immigration enforcement. The second bad option is per-community customization: a bespoke robot for each neighborhood, which is unaffordable, does not scale past a pilot, and puts designers in the position of deciding what each community is like. Local legibility is the middle path, and it is available because of the structure we found. The *questions* are shared—who operates this, is it watching me, does it belong here—so a designer does not need to discover them anew for each site. What has to be local is the *answer*: the attribution that a particular street will recognize, in a language it reads, sited where the work is actually needed. One robot, one small stable set of questions, and an answer fitted to the place. That is a tractable design problem in a way that either extreme is not.

6.3 Design Implications

Four moves follow from the findings, each tied to specific evidence. We call them implications rather than guidelines because the strongest form of each would require behavioral corroboration we do not yet have; we say where that gap sits.

Attribute the operator, do not explain the system. The fastest change of mind we recorded came from telling a frightened participant that a person was driving the robot (§5.1). He was not told what it senses, how it navigates, or

where the data goes. The interpretive problem an unmarked machine poses is a problem of agency—somebody put this here—and it is resolved by naming the somebody. Practically this favors visible institutional attribution over capability disclosure: whose robot this is, at a glance, in the place where the question is being asked.

Make the identification answer the local question, not the generic one. Participants repeatedly told us that an unmarked bin is unusable as well as unreadable: “*I wouldn’t even know if I should put my trash in there or not.*” But identification is not a single design. At Starlight Park the participant’s own remedy for a camera that reads as surveillance was not to remove the camera but to make the machine local—bilingual text, decoration the community would recognize—because the question there was whose enforcement this might be. The same label would be inert in a setting where that question does not arise. Identification should be specified per site even when the hardware is not.

Treat siting as a distributive statement. Participants did not evaluate the robot only as users; several evaluated it as allocators. One redirected it, unprompted, from his own well-served block to streets he named as neglected. Where a robot is deployed is therefore read as a claim about which places warrant service, and a pilot that goes to the most convenient or most photogenic neighborhood is making that claim whether or not it intends to.

Motion carries intent; jerk destroys it. Movement is what makes the object a question at all—a robot that lay tipped over for thirteen minutes drew no recorded attention whatever (§3.1). But motion quality then governs how the question is answered. A participant with professional experience of industrial robots put it directly: acceptance comes down to smoothness, because “*when you see it and jerks, then I think people are like, oh my God, is it coming towards me or not.*” Legibility of intent is thus a precondition for the interpretive work, not a substitute for it.

An open question: approach distance. Participants volunteered specific comfortable distances—roughly two, five, six and ten feet—and grounded them in different things: hygiene, being filmed, being approached without having invited it. This is the most promising candidate for a genuine guideline, because it is behaviorally checkable: the onboard footage records how close the robot actually came and what people did about it. That comparison has not been made, and until it is we report the stated preferences as evidence about what people want to control rather than as a recommended figure. *Note: Retitled from “Design Guidelines” to “Design Implications”. The title previously committed to named, evidence-linked artifacts; four of the five items here are grounded in specific cases, but none is behaviorally corroborated in the strong sense. If someone measures observed approach distances against the stated preferences, the last item becomes a real guideline and the section can revert to the stronger title.*

7 LIMITATIONS

The robot was not autonomous, and the presentation was not uniform. Participants reasoned about a machine they were given to understand was self-directed; it was driven. This is standard for Wizard-of-Oz probe work and it is what allowed us to hold the machine constant across six sites, but it means that accounts of “autonomy” in our corpus are accounts of a presentation rather than of a system, and that the operator is part of the deployment whether or not participants can see them [38]. The presentation was also not held constant. In at least one Bronx interview the operator was disclosed when a participant asked directly, while in another interview in the same session autonomy was asserted; these inconsistencies are recorded in our deviation reports to the IRB, and we flag them in the affected cases (§5.1) rather than smoothing them out. The concealment failed in the other direction as well. At least three participants across two Staten Island sessions reached the operator without being told: one said he had assumed the researchers were directing the robot toward him, one asked whether it would still work if the operator were controlling it from

indoors, and one described having seen somebody operating a garbage can with cameras. Wizard-of-Oz concealment is an assumption about what bystanders can infer, and in a public setting where the wizard has to stand somewhere, it is an assumption that fails at a non-trivial rate. Where it failed, what participants then reasoned about was a teleoperated machine, and we read those accounts accordingly. *Note: RESTORED 2026-08-30 (Xiang). This paragraph existed until 8-29 and was removed with the Method consent rewrite. It has to be here in some form, because §5.1 now states that the trajectory case turns on a disclosure that departed from protocol. Chenchen / PI: please check the wording against the deviation report before submission, and cut the second paragraph if the team would rather not foreground the detection cases.*

This is a demonstration, not an inference. We present cases to show that a practice occurs and what it draws on. We do not claim that neighborhoods differ in measurable ways, that any site produces any response at a rate, or that our participants represent their communities. Because setting type, borough, demographics and existing sanitation infrastructure covary completely across our sites (§3.3), a site-level effect could not be identified here even in principle, and none is asserted.

Coverage is uneven across sites and within them. The study spans all five boroughs, but it remains a single city, in a single country, with a specific sanitation regime and a specific politics of street surveillance; we would expect the practice to generalize and the resources not to. The six sites were also not collected together: the 2025 deployments and the 2026 deployments are roughly a year apart. Within sites, the paired analysis is constrained by two separate limits rather than one. the barrel’s rim is in frame for only part of each recording (§5.1), so the number of interviews we can pair to a documented act is substantially smaller than the number for which footage nominally exists. We therefore present paired cases as cases, and make no claim about what proportion of the corpus they represent.

Participant counts are counts of interviews. Several participant identifiers cover pairs or groups who were interviewed together. The number of people who spoke to us is therefore somewhat higher than the 89 interview records in Table 1, and we do not report a person-level n because we cannot construct one reliably from the recordings.

Recording consent. Participants were asked for permission to record, and some set limits on it. One participant declined both audio and video; their interview and all associated footage are excluded from the corpus and they are not analyzed or quoted anywhere in this paper. Where a participant declined video but agreed to audio, we retain the material under that permission.

Transcription, and what it did to non-English speech. Verbatim material is verified against audio (§4.1), but verification is complete only for the material quoted here; the wider corpus retains uncorrected ASR. One consequence deserves stating rather than burying. Our transcription model reports a detected language per recording, and it returned English for every recording in the corpus—including the interviews in which participants were plainly not speaking English. Rather than flagging speech it could not handle, the model rendered it as approximate English built from the phonetics, so that Spanish passages surface as fluent-looking but meaningless English lines. These passages are not transcripts and we do not quote them; the accounts of the participants concerned reach us only through what an interpreter said on their behalf, or through analyst fieldnotes. The effect is that the corpus systematically under-represents exactly the participants for whom the robot’s lack of any language was most consequential, and it does so invisibly, which is the part worth warning others about. *Note: Verified 2026-08-21 by re-reading the ASR JSON: language is en for all 73 recordings. IMG_1745 (AP-P02) is the clearest case — the deck records it as a Spanish-language interview, and*

the transcript is English gibberish with one Spanish word (“basura”) surviving. IMG_1883 (part of AP-P16) is close to unusable, consisting largely of a repeated “Why?”. Consider whether this belongs in Limitations or is worth a short methods note of its own.

8 CONCLUSION

We used a Wizard of Oz methodology to deploy a trash barrel robot across six structurally different sites spanning all five boroughs of New York City, and interviewed people to understand how they made sense of it. The answers were both the same everywhere and different everywhere. The interpretive move did not vary: people resolved the machine’s ambiguity by reading it through the order of the place it had entered, asking who runs it here, whether it is watching them, and whether it belongs. But there were variances in response, and these can be attributed to the pre-existing awareness locals have about each place—a sanitation worker’s knowledge of her own route, a block already swept every morning by porters, a lawn where people lie in swimsuits, a corner with a reputation.

Roboticians and robot operators are interested in how robots are perceived, but our research illustrates how citizens tend to perceive robots through the their knowledge of the community; what populations, services, issues and routines are typical of the environs. Thus, to anticipate the reception robots will receive in a community, it is important for roboticists to develop a situated understanding of the community, or test the robots in the community in a way that enables them to understand the nuanced and situated responses that the robots elicit.

The same features of the robot acquired different meanings across these settings. At Astoria Park in Queens, participants connected its camera to concerns about filming people sunbathing; at Starlight Park in the Bronx, one interview raised the possibility that the camera would be associated with policing or immigration enforcement. Assessments of the robot’s service likewise drew on local arrangements. In the Bronx’s Little Italy, participants evaluated its usefulness against the work of existing sanitation crews and proposed redirecting it to streets they considered less well served. Local institutions and reputations also informed judgments about the robot’s provenance and safety. At the Roosevelt Island café in Manhattan, the university setting supported interpretations of the robot as a student project. At Tappen Park on Staten Island, participants drew on the park’s reputation to anticipate theft and recommend daytime operation. At the Brooklyn Navy Yard, accounts distinguished the robot’s safety within the controlled campus from anticipated risks on the streets beyond it.

Across these differences, we identify a recurring *situated sensemaking practice*: people interpreted the robot by relating it to the institutions, activities, and expectations they associated with the surrounding place. Questions about responsibility, recording, usefulness, and belonging recurred across settings; the knowledge people used to address them was locally specific. The comparison thus reveals how different judgments can emerge through similar interpretive work. Linked encounters further showed that recognizing and using the robot did not necessarily entail endorsing its deployment. A robot could be readily understood as a trash can while its presence in that place remained open to question.

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